25901680 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCFTQ100 - FOOD ADDITIVES

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **1** marks

1. In which fruits and vegetables is lycopene predominantly found? [U] / [CO5]
2. Which metal ion can catalyze oxidation in food products? [K] / [CO4]
3. What is the primary use of aspartame in food products? [U] / [CO5]
4. What makes honey different from other natural sweeteners? [K] / [CO5]
5. What is the primary function of hydrocolloids in food? [U] / [CO5]
6. What is one method commonly used for flavour encapsulation? [K] / [CO5]
7. What is the function of bulking agents in food? [U] / [CO3]

[1x5 = 5]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. How do preservatives help extend the shelf life of food? [K] / [CO1]

9. What is the definition of food preservatives and what is their purpose in food? [K] / [CO3]

10. How do fungicidal agents function in preventing fungal growth? [U] / [CO2]

11. Explain the role of additives in producing sugar-free or low-fat alternatives. [K] / [CO5]

12. What is the function of leavening agents in baking? [K] / [CO5]

13. Discuss the different types of emulsifiers used in food products. [K] / [CO5]

14. What are the methods used for extracting natural colorants from plants? [K] / [CO5]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **1** questions

Each question carries **10** marks

15. Discuss the role of hydrocolloids, leavening agents and pH control agents in food industry. [K] / [CO5]

16. Describe the function of buffering salts and anticaking agents in food processing. How do they help maintain the stability and quality of food products? [K] / [CO5]

17. How do synthetic antioxidants prevent food oxidation, and what are the pros and cons of using them in food processing? [K] / [CO5]

[10x1 = 10]